Multi-type Facility Location for Urban Drone Logistics with Pickup-and-delivery Demand

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Abstract

Urban low-altitude logistics has experienced rapid growth in recent years and has become a significant supplement and alternative to ground transportation. How to reasonably deploy drone sites and form an efficient network at the urban scale is the key to the healthy development of low-altitude logistics. However, existing studies generally do not distinguish the sub-functions of the sites and lack flexibility in site configuration. To address these challenges, this research introduces a multi-type facility location framework tailored to pickup-and-delivery demand scenarios, such as takeaway and same-city express delivery. We divide the drone stations into four types: pickup station, delivery station, battery exchange station, and parking station. A mixed-integer programming model is proposed to jointly optimize the location, quantity, capacity, and connectivity of each station type. The model is grounded in real operational data from Shenzhen, China. This study expects to make a theoretical breakthrough in the strategic-level optimization problem of low-altitude logistics network planning, thus improving the forward-looking and global optimality of the drone facility deployment.

Keywords: Network design, Drone-based delivery, Hierarchical Facility Location

1. Introduction

The rapid expansion of e-commerce and the corresponding surge in consumer demand for faster delivery times have placed immense pressure on last-mile logistics. Traditional ground-based delivery networks are increasingly constrained by urban traffic congestion, leading to higher operational costs and longer delivery windows. In response, Unmanned Aerial Vehicles (UAVs), commonly known as drones, have emerged as a transformative technology with the potential to revolutionize last-mile delivery. The global drone logistics and transportation market is projected to experience explosive growth, with some forecasts predicting a compound annual growth rate (CAGR) of over 50% to reach a market size of over USD 82 billion by 2034. By operating in low-altitude airspace, drones can bypass ground congestion, offering significant reductions in delivery time, operational costs, and carbon emissions.

This technological shift is particularly salient in the on-demand food delivery sector, a key segment of the broader Urban Air Mobility (UAM) vision. Companies are actively developing and deploying drone-based solutions to deliver meals quickly while preserving food quality. A prominent example is Meituan, one of China's largest on-demand service platforms, which has been operating drone delivery services in dense urban centers like Shenzhen since 2021. Having completed hundreds of thousands of orders with an average delivery time of approximately 15 minutes, Meituan's operations demonstrate the viability and efficiency of a well-designed drone logistics network. This network relies on a sophisticated infrastructure of automated airports (drone hubs) and drop-off lockers to facilitate its service.

The success of such large-scale operations hinges on the strategic design of the underlying infrastructure. This presents a complex strategic decision problem: how to optimally locate and configure a multi-echelon network of facilities to serve a distributed, user-to-user (U2U) demand pattern, characteristic of the food delivery market. **?** This network must comprise several distinct facility types: Drone Bases, where drones are housed and maintained; Pickup Stations (e.g., restaurants); Delivery Stations (e.g., residential or office building collection points); and Recharging Stations to extend operational range. A critical operational

feature in this context is the possibility of "consecutive deliveries," where a drone, upon completing a delivery, may proceed directly to a nearby pickup station for its next task rather than returning to its home base. This behavior creates a flow-dependent system where the availability and routing of drones are influenced by the real-time distribution of demand. This research addresses the challenge of designing this four-echelon infrastructure by developing a comprehensive mathematical model that captures these unique operational characteristics.

The problem studied herein is rooted in the rich literature of the Facility Location Problem (FLP), a cornerstone of operations research concerned with optimally placing facilities to serve customers. Given the multi-echelon nature of our system, our work is more specifically related to the Hierarchical Facility Location Problem (HFLP), which considers interdependencies between different levels of facilities. While HFLP has been applied to logistics and drone delivery, existing models typically focus on simpler two- or three-level systems, such as locating launch and recharge stations, and do not capture the complete four-echelon structure required for a U2U ecosystem. Research on facility location for drone networks has explored various facets, but a comprehensive model integrating these four specific facility types for a U2U market remains an open area.

Two specific streams of literature are particularly relevant to the novel aspects of our model. First, the "consecutive deliveries" feature introduces a dynamic flow of resources (drones) within the network. This structure bears a conceptual resemblance to Closed-Loop Supply Chain (CLSC) design, where reverse logistics models are used to manage the flow of returned products from customers back to collection or remanufacturing centers. In our model, the decision of whether a drone returns to a base or proceeds to a new pickup point can be viewed as managing a "return" flow of available assets to satisfy new demand. Second, the placement of recharging stations is related to the Refueling Station Location Problem. Our approach is particularly inspired by flow-capturing location models (FCLM), which aim to locate facilities to maximize the volume of traffic flow they can service along paths. However, unlike some FRLM variants that model complex multi-stop refueling paths, we adopt a simplified assumption that a drone will not visit multiple recharging stations during a single delivery leg, reflecting the operational preference of companies like Meituan for time-sensitive deliveries.

From a methodological standpoint, the resulting facility location model is a large-scale, NP-hard mixed-integer linear program (MILP). Benders decomposition is a classical and powerful exact method for solving such problems, partitioning them into a master problem that handles strategic location decisions and a subproblem that solves the resulting operational flow allocation. Despite its theoretical elegance, the standard algorithm can suffer from slow convergence, often necessitating acceleration techniques such as the generation of stronger cuts. As a complementary approach, Lagrangian relaxation is another widely used decomposition technique in facility location, which relaxes complicating constraints to create an easier subproblem that provides a strong bound on the optimal solution and can form the basis of effective heuristics. Finally, to model the non-linear relationship between the inbound drone flow at a delivery station and the proportion available for consecutive deliveries, we employ a standard operations research technique: piecewise linear approximation. This allows us to capture this complex, flow-dependent behavior within the tractable framework of an MILP.

We contributing as follows:

- *A Novel Flow-Dependent Hierarchical Formulation*: We introduce a new maximum coverage-based hierarchical facility location model specifically designed for the user-to-user (U2U) demand patterns in urban delivery systems. To our knowledge, this is the first study that simultaneously determines the location of four-echelon drone infrastructures, including drone bases, pickup stations, delivery stations, and recharging stations. A significant innovation is the explicit modeling of operational dynamics arising from consecutive deliveries, where drones completing a delivery may either return to the bases or proceed to the next pickup station. The model introduces a flow-dependent proportion to represent this behavior, which is linearized through a piecewise function to capture the non-linear relationship between inbound drone flow and consecutive delivery decisions.

- *Tailored Solution Methods*: To solve the proposed large-scale, NP-hard optimization problem, we develop a specialized exact algorithm based on Benders decomposition. The computational efficiency

of this framework is significantly enhanced by integrating tailored acceleration strategies, including the generation of Pareto-optimal cuts and the use of strong heuristics for warm-starting the procedure. Furthermore, we formulate a Lagrangian relaxation of the problem, which not only provides tight lower bounds for benchmarking but also lays the groundwork for developing fast, high-quality heuristics in future work.

- *Pragmatic Strategic Insights*: Preliminary computational experiments confirm the effectiveness of the accelerated Benders decomposition method, which demonstrates a substantial reduction in solution time compared to standard commercial solvers. The model provides a robust quantitative tool for strategic decision-making, enabling managers to analyze the trade-offs between capital investment in infrastructure and the resulting market coverage...

The remainder of this paper is organized as follows. Section **??** provides a review of related literature. In Section 2.4, we describe the loading procedures and also explain the difficulties faced in practice loading. Section **??** utilizes the Benders decomposition method as well as the Lagrangian decomposition. The computational results are presented in Section **??** to demonstrate the computational efficiency, followed by the concluding remarks and future research directions in Section **??**.

2. Literature Review

Our research intersects three main streams of literature within operations research and supply chain management: network design for low-altitude logistics, hierarchical facility location with a focus on charging infrastructure, and facility location models incorporating congestion and queueing effects.

2.1. Network Design for Low-Altitude Logistics

The integration of unmanned aerial vehicles (UAVs), or drones, into commercial and emergency logistics has spurred a new domain of network design focused on low-altitude operations. Due to the limited flight range of battery-powered drones, multiple recharging stations are required in large urban areas to complete deliveries without running out of power (Hong et al., 2017). Consequently, the optimal placement of these bases and the routing of drones heavily depend on continuous space shortest paths and range-restricted flow-refueling logic (Hong et al., 2018).

Recently, drone networks have been increasingly deployed to deliver time-critical materials, such as medication, blood, and exam kits, in emergency situations (Wang et al., 2022). In this context, Boutilier and Chan (2022) developed an integrated location-queuing model based on a p-median architecture to design a defibrillator-enabled drone network that augments existing emergency medical services. Similarly, Lejeune and Ma (2025) proposed a stochastic emergency network design model for opioid overdose response, formulating the network as a collection of M/G/K queueing systems where the capacity of each system is an endogenous decision variable. Because the resulting nonlinear integer problem is intractable in its original form, they developed a mixed-integer linear programming (MILP) reformulation and a branch-and-cut algorithmic framework.

To account for system congestion in drone-based emergency delivery, Wang et al. (2022) jointly optimized the location of facilities, the capacity of drones at opened facilities, and the allocation of demands. They employed queues to model the system congestion of drone requests under non-priority, static priority, and dynamic priority disciplines, subsequently approximating the model as a computationally tractable mixed-integer second-order conic program (MISOCP).

Beyond emergency logistics, Chen et al. (2025) recently introduced a coordinated drone-courier logistics system for intra-city express services, recognizing that drones are often unable to execute "last-meter" deliveries directly to customer doorsteps due to airspace and safety restrictions. In their system, drones operate within a closed network among vertiports, while couriers connect customers to the drone delivery system. They formulated an integrated optimization model where the interplay between network planning and tactical operations is mirrored by a queueing network model. To handle the resulting nonlinear and partially convex/concave feasible region, they developed an iterative exact algorithm that adaptively refines

linear approximations of the nonlinear constraints. Our work extends this stream by explicitly capturing chaining tasks and unified energy balances across a multi-type drone network.

2.2. Hierarchical Facility Location and Charging Infrastructure

The second stream of relevant literature encompasses hierarchical facility location problems (HFLP) and the optimal deployment of alternative fuel and charging infrastructure. The primary objective in a typical hierarchical facility location problem is to determine the location of facilities in a multi-level network in a way to serve the customers at the lowest level of the hierarchy both efficiently and effectively (Farahani et al., 2014).

With the electrification of transportation, a significant portion of this literature has pivoted toward the charging station location problem (CSLP), which seeks to optimize the locations of charging stations to satisfy customer recharging demand under real-life restrictions (Kchaou-Boujelben, 2021). A widely adopted paradigm is the flow-refueling location model (FRLM), which positions refueling facilities to maximize the flow that can be accommodated in a road network while respecting the range limitations of vehicles (MirHassani and Ebrazi, 2013). Extensions of the FRLM have explicitly added the routing aspect of individual drivers and their shortest path deviation tolerances (Yıldız et al., 2016).

However, charging an electric vehicle (EV) is inherently a time-consuming burden, which can lead to severe congestion if the public charging infrastructure is not well-designed (Li et al., 2025). Addressing this, Li et al. (2025) studied the strategic planning of public EV charging stations under uncertain charging demand and service rates, introducing chance constraints to mitigate long waiting times.

A major paradigm shift in mitigating charging times is the joint deployment of standard chargers and battery swapping stations. Lai and Li (2024) proposed a multimodal charging network in which charging stations and battery swapping stations are jointly built to synergistically support an electric ride-hailing fleet. Their rationale is that while charging stations are cost-effective for scaling up fleets, battery swapping stations offer quick turnaround and can improve fleet utilization (Lai and Li, 2024). Incorporating elastic passenger demand and spatial charging equilibrium, they formulated the problem as a nonconcave program and established tight upper bounds through relaxation and decomposition (Lai and Li, 2024).

Furthermore, the operational integration of charging infrastructure within vehicle sharing systems is heavily explored. He et al. (2020) investigated how to charge an EV-sharing fleet to ensure viability and profitability by integrating charging infrastructure planning and vehicle repositioning operations. They developed a queuing network model that characterizes how customers endogenously pick EVs based on energy levels and how the operator implements charging policies. Similarly, He et al. (2017) formulated a mathematical programming model for service region design that incorporates customer adoption behavior and fleet management, addressing inherent planning uncertainty through a distributionally robust optimization framework approximated by an MISOCP. Our model draws inspiration from these hierarchical and operational frameworks but applies them to the unique requirements of drone base and battery swapping station co-location.

2.3. Facility Location with Congestion and Queueing Effects

The traditional facility location literature often assumes infinite processing capacity at facilities, ignoring the waiting times incurred by users. However, in service networks, the facility location problem with immobile servers, stochastic demand, and congestion is highly critical, often modeled as a network of M/M/1 queues leading to complex nonlinear mixed-integer programs (Elhedhli, 2006).

To model the flow of entities in constrained logistics networks, researchers have utilized closed queueing networks. For instance, Adelman (2007) considered an internal pricing mechanism for managing a fleet of service units flowing in a closed logistics queueing network, utilizing Palm calculus to approximate queueing and stockout effects. Shu et al. (2013) developed a network flow model with proportionality constraints to estimate the flow of bicycles within a public bicycle-sharing system, given an initial allocation of bicycles. He et al. (2017) developed closed queue network for electric vehicle sharing systems, and the ideas are extended to air mobility network (Kai et al., 2022; Wang et al., 2025) and intra-city express services (Chen et al., 2025).

The primary mathematical challenge in these congestion models lies in handling the nonlinear, fractional queueing formulas within integer programming frameworks. While earlier works relied on piecewise linear approximations (Elhedhli, 2006), recent methodological advancements have proven that mixed-integer second-order cone programs (MISOCPs) form a highly efficient class of mixed-integer convex programs for these problems (Ahmadi-Javid and Hoseinpour, 2022). Ahmadi-Javid and Hoseinpour (2022) comprehensively demonstrated how various performance metrics of M/G/1 queues can be modeled by different MISOCPs. They showed that conic reformulation methods efficiently solve very large-size test problems that cannot be solved by existing methods based on linear programming or state-space decomposition within reasonable times.

This SOCP convexification approach has become the state-of-the-art technique in recent OR/MSOM literature. For example, He et al. (2020) utilized lower and upper bound formulations as mixed-integer second-order cone programs to achieve computationally tractable solutions for their integrated queuing-location EV model. Wang et al. (2022) similarly leveraged MISOCP reformulations to solve their drone facility location model with congestion and priority queuing. Aligning with this advanced methodological stream, our work reformulates the highly non-linear fractional terms associated with drone waiting times and base congestion into an SOCP. This allows us to rigorously incorporate congestion effects and chaining logic while maintaining computational tractability using off-the-shelf solvers.

2.4. Problem Description

We consider a multi-type drone network design problem for urban low-altitude logistics. The network operates over a partially connected graph comprising four primary types of nodes: *candidate drone bases* $\mathcal{L}$, *pickup stations* $\mathcal{P}$, *delivery stations* $\mathcal{D}$, and *battery swapping stations* $\mathcal{E}$. Drones provide point-to-point delivery services to fulfill orders occurring between origin-destination (O-D) pairs.

Similar to the works by Adelman (2007); He et al. (2017); Chen et al. (2025), we model the drone fleet operations as a closed queueing network. A key feature to note in our modeling is that the drones, rather than the orders or the packages, are the active entities that flow within the queueing network. In a closed network, the total number of operational drones is conserved. At any given time instant, a drone can be in one of the following states: (i) being idle at a physical drone base awaiting the next assigned order; (ii) executing a *loaded flight*, i.e., traveling from a pickup node to a delivery node with an order; (iii) engaging in *deadhead repositioning*, such as traveling from a base to a pickup node, or returning from a delivery node to a base; or (iv) queuing or receiving service for battery replenishment at a charging or swapping facility. Correspondingly, we define various types of queues to characterize the time delays and congestion drones undergo during these activities. In the following discussion, we use the terms "queue" and "node" interchangeably, as both refer to individual service stations in the queueing network.

To significantly improve the turnover efficiency of the drone fleet, our system incorporates the logic of *chaining tasks*. After completing a delivery at a delivery station, a drone is not strictly required to return to a physical base. Instead, it may be directly dispatched to a nearby pickup station to serve a new order. To mathematically capture this chaining mechanism within the queueing framework, we introduce a set of *virtual bases* $\mathcal{L}^{\mathrm{v}}$, each corresponding exactly to a delivery node. A drone entering a virtual base implies it is temporarily idling near the delivery point, waiting for a chaining task, subject to a maximum allowable waiting time constraint.

Furthermore, drones are heavily constrained by their *battery capacity*. They consume energy uniformly during both in-transit flights and while idling in queues. To maintain a global *energy balance*, drones with depleted batteries must detour to battery swapping stations or return to physical bases to restore their energy to a safe usable capacity before continuing operations.

Modeling assumptions. We structure our model upon three practical assumptions that align with real-world operational environments:

- The deployment of drones primarily substitutes human couriers in intra-city express services, leading to significantly reduced delivery times and lower marginal costs.

- The individual utilities or dynamic preferences of customers are not explicitly modeled; instead, the system's decision-making is driven by strategic, system-level service capacity and accessibility.
- In high-demand areas or during peak hours, it is infeasible to assign drones to all incoming orders due to congestion and fleet capacity limits. Thus, our model focuses on maximizing the total fulfilled orders (i.e., coverage of potential drone-service demand) under strict budget and fleet size constraints.

Modeling assumptions. We first restate three practical assumptions that shape the structure of our model:

- The deployment of drones primarily substitutes human couriers, leading to reduced delivery time and lower marginal cost.
- The customers' individual utilities or preferences are not considered explicitly, as the decision-making is driven by system-level service capacity and accessibility.
- In high-demand areas or during peak hours, it is infeasible to assign drones to all incoming orders. Thus, the model focuses on maximizing the coverage of potential drone-service areas under limited operational capacity.

3. Model Formulation

We formulate the multi-type drone network design problem by considering chaining tasks, congestion effects, and battery constraints. Let $\mathcal{G} = (\mathcal{N}, \mathcal{A})$ be the network graph, where the node set $\mathcal{N} = \mathcal{L} \cup \mathcal{P} \cup \mathcal{D} \cup \mathcal{E}$ consists of candidate drone bases $\mathcal{L}$, pickup stations $\mathcal{P}$, delivery stations $\mathcal{D}$, and battery swapping stations $\mathcal{E}$. The set $\mathcal{A}$ denotes the connected edges in the partially connected graph; accordingly, subsets of arcs are denoted by their respective node sets, e.g., $\mathcal{A}^{PD}$ represents the links (u, v) where $u \in \mathcal{P}$ and $v \in \mathcal{D}$.

To accurately model the chaining delivery logic, we introduce a set of virtual bases $\mathcal{L}^{\mathrm{v}}$, where each virtual base $l' \in \mathcal{L}^{\mathrm{v}}$ corresponds exactly to a delivery node $j \in \mathcal{D}$. The comprehensive set of all bases is thus denoted as $\mathcal{L} = \hat{\mathcal{L}} \cup \mathcal{L}^{\mathrm{v}}$, where $\hat{\mathcal{L}}$ represents the physical bases.

The model is built upon a closed queuing network framework, utilizing drone flow to represent fulfilled orders. Assuming a strict one-to-one drone-to-order ratio (e.g., standard operations in systems like Meituan), our primary objective is to maximize the total number of served demands (fulfilled orders) across the network within a specified budget limit.

Orders arrive between pickup and delivery stations following a Poisson process. We assign all potential Origin-Destination (O-D) pairs to specific drone bases. Drones stationed at a base are dispatched immediately upon the arrival of an assigned order; if no drones are available in the queue, the order is assumed to be lost. To capture this, we define a continuous decision variable ρ, representing the service utility (or service level), which is uniform across all bases and reflects the percentage of orders successfully captured by the drones.

3.1. Order Assignment and Linearization

We first define the logical assignment of orders and flow generation.

ρ_l Continuous variable denoting the service utility.

x_u Binary variable, equal to 1 if node $u \in \mathcal{N}$ is activated, and 0 otherwise.

y_{li} Continuous variable denoting the proportion of orders from pickup node $i \in \mathcal{P}$ assigned to base $l \in \mathcal{L}$.

ψ_{uv} Continuous variable representing the aggregated physical drone flow on link (u, v).

η_{ls} Binary, equals 1 if base l have the service level of s.

The logical assignments and resulting flows are governed by the following constraints:

$$\sum_{l\in\mathcal{L}} y_{li} \le x_i, \qquad \forall i \in \mathcal{P} \tag{1}$$

$$y_{li} \le x_l, \qquad \forall i \in \mathcal{P}, l \in \mathcal{L} \tag{2}$$

$$\sum_{s\in\mathcal{S}} \rho_s \eta_{ls} \cdot y_{li} \cdot \sum_{j:(i,j)\in\mathcal{A}^{PD}} h_{ij} x_j = \psi_{li}, \qquad \forall (l,i) \in \mathcal{A}^{LP} \tag{3}$$

$$\sum_{s\in\mathcal{S}} \eta_{ls} = x_l, \qquad \forall l \in \mathcal{L} \tag{4}$$

Constraints (1) and (2) restrict assignments such that orders are only routed between actively functioning pickup nodes and bases. Constraint (3) maps the potential demand h_{ij} to the actual logical flow ψ_{li}, scaled by the service rate ρ and the assignment variable y_{li}.

Because Constraint (3) contains non-linear multiplications of continuous and binary decision variables ($\eta_{ls} \cdot y_{li} \cdot x_j$), we introduce auxiliary variables to exactly linearize the formulation. First, let $w_{lij} = y_{li} x_j$. Since $y_{li} \in [0,1]$, this product is linearized using the standard Big-M method with $M = 1$. We aggregate the potential demand directed to base l from node i as $D_{li} = \sum_{j:(i,j)\in\mathcal{A}^{PD}} h_{ij} w_{lij}$. Next, we define $z_{lis} = \eta_{ls} D_{li}$. Given that D_{li} is bounded by the maximum possible demand $H_i = \sum_j h_{ij}$, we linearize this product using $M = H_i$. The exact linearized formulation replaces (3) with the following constraints:

$$w_{lij} \le y_{li}, \quad w_{lij} \le x_j, \qquad \forall l \in \mathcal{L}, (i,j) \in \mathcal{A}^{PD} \tag{5}$$

$$w_{lij} \ge y_{li} - (1 - x_j), \quad w_{lij} \ge 0, \qquad \forall l \in \mathcal{L}, (i,j) \in \mathcal{A}^{PD} \tag{6}$$

$$D_{li} = \sum_{j:(i,j)\in\mathcal{A}^{PD}} h_{ij} w_{lij}, \qquad \forall l \in \mathcal{L}, i \in \mathcal{P} \tag{7}$$

$$z_{lis} \le D_{li}, \quad z_{lis} \le H_i \eta_{ls}, \qquad \forall l \in \mathcal{L}, i \in \mathcal{P}, s \in \mathcal{S} \tag{8}$$

$$z_{lis} \ge D_{li} - H_i(1 - \eta_{ls}), \quad z_{lis} \ge 0, \qquad \forall l \in \mathcal{L}, i \in \mathcal{P}, s \in \mathcal{S} \tag{9}$$

$$\psi_{li} = \sum_{s\in\mathcal{S}} \rho_s z_{lis}, \qquad \forall l \in \mathcal{L}, i \in \mathcal{P} \tag{10}$$

3.2. Flow Conservation and Congestion

We bridge the logical flows to physical link flows and explicitly model congestion at pickup and delivery nodes. Upon completing a delivery, drones face a routing choice: return to a physical base or engage in a chaining task. We conceptualize the latter by directing these drones into a virtual base l' associated with the corresponding delivery node.

τ Parameter defining the maximum allowable waiting factor at virtual bases.

$$\sum_{u:(u,v)\in\mathcal{A}} \psi_{uv} = \sum_{u:(v,u)\in\mathcal{A}} \psi_{vu}, \qquad \forall v \in \mathcal{N} \tag{11}$$

$$\psi_{ij} \le h_{ij}, \qquad \forall (i,j) \in \mathcal{A}^{PD} \tag{12}$$

$$\mathbb{P}(W_\ell > \tau) \le \gamma \qquad \forall \ell \in \mathcal{L}^V \tag{13}$$

$$\sum_{s\in\mathcal{S}} \frac{\rho_s}{1-\rho_s} u_{\ell s} \le \tau \sum_{i:(\ell,i)\in\mathcal{A}^{L^V P}} \psi_{\ell i}, \forall \ell \in \mathcal{L}^{\mathrm{v}}$$

Constraint (11) ensures physical flow conservation across the network nodes. Specifically at delivery nodes, arriving drones must subsequently flow either to a physical base or a virtual base. Constraint (12) ensures the actual flow does not exceed the potential demand. Constraint (13) bounds the average waiting time at virtual bases to maintain efficient turnover for chaining tasks.

3.3. Battery Swapping and Energy Balance

To accommodate operational battery limitations, we decompose the total aggregated flow into direct flights and detoured flights via swapping stations. We enforce a global energy balance across the network. While physical drone bases can inherently charge drones, we isolate the swapping logic by formally separating bases and swapping stations. For notational convenience, we define $\hat{\mathcal{E}} = \mathcal{E} \cup \{0\}$, where index 0 represents a direct flight (no swapping).

φ_{uv}^k Continuous variable, the flow component on link (u, v) routing via swapping station k.

ω, Ω Parameters denoting the energy consumption rate and the maximum battery capacity, respectively.

t_{uv}^k Parameter representing the travel time on link (u, v) via swapping station k (where t_{uv}^0 is the direct flight time).

$$\psi_{uv} = \sum_{k \in \hat{\mathcal{E}}_{uv}} \varphi_{uv}^k \qquad \forall (u,v) \in \mathcal{A} \tag{14}$$

$$\omega \underbrace{\Big(\sum_{\ell \in \mathcal{L}^{\mathrm{v}}} \sum_{s \in \mathcal{S}} \frac{\rho_s}{1-\rho_s} u_{\ell s} \Big)}_{\text{Virtual base idling}} + \underbrace{\sum_{u \in \mathcal{P} \cup \mathcal{D}} \frac{\sum_v \psi_{uv}}{\mu_u - \sum_v \psi_{uv}}}_{\text{Queuing at P/D nodes}} + \underbrace{\sum_{(u,v) \in \mathcal{A}} \sum_{k \in \mathcal{E}_{uv}} \varphi_{uv}^k t_{uv}^k \Big)}_{\text{In-transit flight}}$$

$$\leq (1-\varepsilon)\Omega \underbrace{\left(\sum_{(u,v) \in \mathcal{A}} \sum_{k \in \tilde{\mathcal{E}}_{uv}} \varphi_{uv}^k + \sum_{(j,l) \in \mathcal{A}^{DL}} \varphi_{jl}^0 \right)}_{\text{Energy replenishments}} \tag{15}$$

$$\omega \Big(\sum_{s \in \mathcal{S}} \frac{\rho_s}{1-\rho_s} u_{\ell s} + \sum_{(\ell,i) \in \mathcal{A}^{LVP}} \sum_{k \in \mathcal{E}_{\ell i}} \varphi_{\ell i}^k t_{\ell i}^k \Big) \leq (1-\varepsilon)\Omega \left(\sum_{(\ell,i) \in \mathcal{A}^{LVP}} \sum_{k \in \tilde{\mathcal{E}}_{\ell i}} \varphi_{\ell i}^k \right) \qquad \forall \ell \in \mathcal{L}^V \tag{16}$$

$$\omega \Big(\sum_{i \in \mathcal{P}} \frac{\sum_j \psi_{ij}}{\mu_i - \sum_j \psi_{ij}} + \sum_{(i,j) \in \mathcal{A}^{PD}} \sum_{k \in \mathcal{E}_{ij}} \varphi_{uv}^k t_{uv}^k \Big) \leq (1-\varepsilon)\Omega \left(\sum_{(i,j) \in \mathcal{A}^{PD}} \sum_{k \in \tilde{\mathcal{E}}_{ij}} \varphi_{ij}^k \right) \qquad \forall i \in \mathcal{P}$$

$$\omega \Big(\sum_{j \in \mathcal{D}} \frac{\sum_l \psi_{il}}{\mu_j - \sum_l \psi_{jl}} + \sum_{(j,l) \in \mathcal{A}^{DL}} \sum_{k \in \mathcal{E}_{jl}} \varphi_{jl}^k t_{jl}^k \Big) \leq (1-\varepsilon)\Omega \left(\sum_{(j,\ell) \in \mathcal{A}^{DLV}} \sum_{k \in \tilde{\mathcal{E}}_{j\ell}} \varphi_{j\ell}^k + \sum_{(j,l) \in \mathcal{A}^{D\tilde{L}}} \varphi_{jl}^0 \right) \qquad \forall j \in \mathcal{D}$$

Constraint (14) partitions the physical flow into its swapping and non-swapping components. Constraint (15) characterizes the system-wide energy balance. The left-hand side calculates the total energy consumption rate by multiplying the uniform power consumption rate (ω) with the total number of active drones in the system. The drones are distributed across three states: (1) idling at virtual bases waiting for chaining tasks, evaluated by the M/M/1 queue length approximation $\frac{\rho}{1-\rho}$; (2) waiting in queues or receiving service at pickup and delivery stations; and (3) in transit on the flight links, calculated by Little's Law as the product of flow rate φ_{uv}^k and travel time t_{uv}^k. The right-hand side represents the total energy replenishment rate, assuming each visit to a swapping station or a physical base restores the battery to a safe usable capacity of $(1-\varepsilon)\Omega$.

3.4. Service Quality, Fleet Sizing, and Budget Constraints

We guarantee the Quality of Service (QoS) by imposing probabilistic queue length constraints. Let $\lambda_u = \sum_v \psi_{uv}$ be the total outflow rate at node u.

$\bar{\Lambda}(\mu_u)$ Parameter representing the maximum allowable arrival rate for a given service rate μ_u to satisfy the QoS queue length probability constraint.

Γ Continuous variable for the total required fleet size.

f_u, c Parameters representing the fixed cost of activating node u and the unit acquisition cost of a drone, respectively.

$V, C_{\mathbf{budget}}$ Parameters for the maximum allowable delivery time and the total financial budget.

$$\mathbb{P}(W_u > \tau) \leq \gamma \qquad \forall u \in \mathcal{P} \cup \mathcal{D} \cup \tilde{\mathcal{E}} \tag{17}$$

$$\sum_{v:(u,v)\in\mathcal{A}} \psi_{uv} \leq \bar{\Lambda}(\mu_u) x_u \qquad \forall u \in \mathcal{P} \cup \mathcal{D} \tag{18}$$

$$\sum_{(u,v)\in\mathcal{A}} \varphi_{uv}^k \leq \bar{\Lambda}(\mu_k) x_k \qquad \forall k \in \tilde{\mathcal{E}} \tag{19}$$

Constraints (18) and (19) ensure that the probability of the queue length exceeding a pre-defined threshold remains strictly below an acceptable violation level, mapping this requirement to a maximum allowable throughput $\bar{\Lambda}(\cdot)$.

To determine fleet sizing, we adopt the fixed-population-mean method. This method approximates the required number of entities (drones) in a closed queuing network by evaluating it as an open queuing network. Using Little's Law, the total required fleet size Γ is the sum of the expected number of drones in each part of the network:

$$\underbrace{\sum_{l\in\mathcal{L}} \sum_{s\in\mathcal{S}} \frac{\rho_s}{1-\rho_s} \eta_{ls}}_{\text{Queuing at physical bases}} + \underbrace{\sum_{u\in\mathcal{P}\cup\mathcal{D}} \frac{\lambda_u}{\mu_u - \lambda_u}}_{\text{Queuing at P/D nodes}} + \underbrace{\sum_{(u,v)\in\mathcal{A}} \sum_{k\in\mathcal{E}_{uv}} \varphi_{uv}^k t_{uv}^k}_{\text{In-transit flight}} \leq \Gamma \tag{20}$$

$$\sum_{u\in\mathcal{N}} f_u x_u + c\Gamma \leq C_{\text{budget}} \tag{21}$$

$$\Gamma \leq V \sum_{l\in\tilde{\mathcal{L}}} x_l \tag{22}$$

In Constraint (20), the first term computes the steady-state number of idle drones waiting at physical bases. The second term computes the expected number of drones at pickup and delivery stations (both waiting and being serviced). The third term computes the expected number of drones currently flying along the network links. Constraint (21) prevents the total annualized capital expenditure—encompassing both base activation costs and fleet acquisition costs—from exceeding the allocated budget. Constraint (22) provides a structural upper bound for the fleet size.

3.5. Full Optimization Model

Let ψ_{ij} denote the aggregated physical loaded flow from pickup node i to delivery node j. The full non-linear mixed-integer programming (MINLP) model is summarized as follows:

$$
\begin{aligned}
\max \quad & Z = \sum_{(i,j)\in\mathcal{A}^{PD}} \psi_{ij} && (23)\\
\text{s.t.} \quad & \text{Order assignment: } (1)-(3)\\
& \text{Linearization constraints: } (\textbf{??})-(\textbf{??})\\
& \text{Flow conservation and logic: } (11)-(13)\\
& \text{Energy and routing logic: } (14), (15)\\
& \text{QoS, fleet size, and budget: } (18)-(22)\\
& x_u \in \{0,1\}, \forall u \in \mathcal{N}\\
& \rho, y_{li}, \psi_{uv}, \varphi_{uv}^k, \Gamma \geq 0
\end{aligned}
$$

3.6. Second-Order Cone Programming (SOCP) Reformulation

By discretizing the service utility into defined levels $s \in \mathcal{S}$, the previously non-linear queuing lengths at physical and virtual bases ($\frac{\rho}{1-\rho}$) are elegantly transformed into a linear combination of constants $\sum_{s\in\mathcal{S}} \frac{\rho_s}{1-\rho_s}\eta_{ls}$. However, the original formulation still contains highly non-linear fractional terms regarding the queuing mechanics at pickup and delivery nodes ($\frac{\lambda_u}{\mu_u-\lambda_u}$ in equations (15) and (20)). To maintain computational tractability and allow the use of off-the-shelf solvers, we reformulate these remaining fractions into a Second-Order Cone Program (SOCP).

We introduce an auxiliary continuous variable $\xi_u \geq 0$ to represent the expect queueing drones at node $u \in \mathcal{P} \cup \mathcal{D}$, bounded by $\xi_u \geq \frac{\lambda_u}{\mu_u-\lambda_u}$. This inequality can be mathematically rearranged into a standard rotated second-order cone constraint:

$$
\left\| \begin{pmatrix} 2\sqrt{\mu_u} \\ \xi_u + \lambda_u - \mu_u + 1 \end{pmatrix} \right\|_2 \leq \xi_u - \lambda_u + \mu_u + 1, \qquad \forall u \in \mathcal{P} \cup \mathcal{D} \tag{24}
$$

Using the newly defined auxiliary ξ_u and the linear discrete service terms, the previously non-linear energy and fleet constraints (15) and (20) are seamlessly reformulated as linear inequalities:

$$
\begin{aligned}
& \omega\Big(\sum_{\ell\in\mathcal{L}^{\mathrm{v}}} \sum_{s\in\mathcal{S}} \frac{\rho_s}{1-\rho_s} u_{\ell s} + \sum_{u\in\mathcal{P}\cup\mathcal{D}} \xi_u + \sum_{(u,v)\in\mathcal{A}} \sum_{k\in\mathcal{E}_{uv}} \varphi_{uv}^k t_{uv}^k \Big) \\
& \leq (1-\varepsilon)\Omega \left(\sum_{(u,v)\in\mathcal{A}} \sum_{k\in\tilde{\mathcal{E}}_{uv}} \psi_{uv}^k + \sum_{(j,l)\in\mathcal{A}^{DL}} \psi_{jl}^0 \right) && (25)\\
& \sum_{l\in\mathcal{L}} \sum_{s\in\mathcal{S}} \frac{\rho_s}{1-\rho_s} \eta_{ls} + \sum_{u\in\mathcal{P}\cup\mathcal{D}} \xi_u + \sum_{(u,v)\in\mathcal{A}} \sum_{k\in\mathcal{E}_{uv}} \varphi_{uv}^k t_{uv}^k \leq \Gamma && (26)
\end{aligned}
$$

4. Solution methods

For fixed values of the binary variables z and u provided by the master problem, the subproblem optimizes all continuous flow and auxiliary variables. The subproblem can be formulated as:

By fixing the binaries, the subproblem is a SOCP problem:

The subproblem is a linear program conditioned on the master variables, which allows generating optimality cuts or feasibility cuts that are added to the master problem iteratively. This decomposition exploits the separable structure of the network and the piecewise-linear approximation, significantly reducing computational complexity.

5. Computational Results and Managerial Implications

5.1. Algorithm Efficiency

5.2. Managerial Insights

5.2.1. Low budget or show hand?

5.2.2. Distance between P and D

5.2.3. Effect of flow-dependent economies of scale on optimal solutions

1. fixed transition (no congestion) 2. different parameter 3.

As stated earlier, using flow-dependent economies of scale in modeling the transportation costs is one of the extensions in this paper, which makes the problem more computationally challenging compared to the classical hub location problem with flow independent discount scheme. In this section, we focus on a comparison analysis to highlight the differences between the optimal solutions of the problem under both discount schemes in terms of network structure. To do this, we solved the problem in its classical setting by applying a fixed discount factor to inter-hub transportation costs. To have a fair comparison, we set the value of discount factor equal to average of the four line segments for each cost function (i.e., 0.85, 0.7, and 0.5 for F1, F2, and F3, respectively). The results for solving the problem with AP25 data set including the optimal value and the optimal set of opened hubs are presented in Table 7. Comparing the results presented in Table 7 with those reported in Table 4, it can be seen that in 18 instances solved from the AP25 data set, the optimal set of hubs differ from that of the problem with flow-dependent discount. In 13 out of these 18 cases, the number of hubs opened under the classical discount exceeds by one the number of hubs opened under flow-dependent discount. In the remaining five cases, the number of opened hubs is the same but their compositions differ. For example, the optimal network configurations for the flow-dependent and fixed discount schemes for the AP25L instance with cost function F2, = 0.75, and = 0.8 are presented in Fig. 4. Since the spoke links (i.e., the allocation decisions) depend on random scenarios, we depict the spoke links corresponding to first scenario. As can be seen, the network configuration in terms of the location of hubs and the established linkages can be different from the case in which the fixed discount setting is used.

5.2.4. Hub-spoke and distributional

comparison between two methods

5.2.5. Number of each stations

1. L or E (which benefit? more bases or more recharging)

5.3. Coverage distance

5.4. Drone utilization

6. Conclusion

References

Adelman, D., 2007. Price-Directed Control of a Closed Logistics Queueing Network. Operations Research 55, 1022–1038.

Ahmadi-Javid, A., Hoseinpour, P., 2022. Convexification of Queueing Formulas by Mixed-Integer Second-Order Cone Programming: An Application to a Discrete Location Problem with Congestion. INFORMS Journal on Computing 34, 2621–2633.

Boutilier, J.J., Chan, T.C.Y., 2022. Drone Network Design for Cardiac Arrest Response. Manufacturing & Service Operations Management 24, 2407–2424.

Chen, S., Wang, K., Wu, L., Qi, W., 2025. On Coordinated Drone-Courier Logistics for Intra-city Express Services. arXiv:2501.05200.

Elhedhli, S., 2006. Service System Design with Immobile Servers, Stochastic Demand, and Congestion. Manufacturing & Service Operations Management 8, 92–97.

Farahani, R.Z., Hekmatfar, M., Fahimnia, B., Kazemzadeh, N., 2014. Hierarchical facility location problem: Models, classifications, techniques, and applications. Computers & Industrial Engineering 68, 104–117.

He, L., Ma, G., Qi, W., Wang, X., 2020. Charging an Electric Vehicle-Sharing Fleet. Manufacturing & Service Operations Management , msom.2019.0851.

He, L., Mak, H.Y., Rong, Y., Shen, Z.J.M., 2017. Service Region Design for Urban Electric Vehicle Sharing Systems. Manufacturing & Service Operations Management 19, 309–327.

Hong, I., Kuby, M., Murray, A., 2017. A Deviation Flow Refueling Location Model for Continuous Space: A Commercial Drone Delivery System for Urban Areas, in: Griffith, D.A., Chun, Y., Dean, D.J. (Eds.), Advances in Geocomputation. Springer International Publishing, Cham, pp. 125–132.

Hong, I., Kuby, M., Murray, A.T., 2018. A range-restricted recharging station coverage model for drone delivery service planning. Transportation Research Part C: Emerging Technologies 90, 198–212.

Kai, W., Jacquillat, A., Vaze, V., 2022. Vertiport Planning for Urban Aerial Mobility: An Adaptive Discretization Approach. Manufacturing & Service Operations Management 24, 3215–3235.

Kchaou-Boujelben, M., 2021. Charging station location problem: A comprehensive review on models and solution approaches. Transportation Research Part C: Emerging Technologies 132, 103376.

Lai, Z., Li, S., 2024. Towards a multimodal charging network: Joint planning of charging stations and battery swapping stations for electrified ride-hailing fleets. Transportation Research Part B: Methodological 183, 102928.

Lejeune, M.A., Ma, W., 2025. Drone-Delivery Network for Opioid Overdose: Nonlinear Integer Queueing-Optimization Models and Methods. Operations Research 73, 86–108.

Li, Y., Shu, J., Wang, C., Wu, T., Wu, Y., 2025. Robust planning for electric vehicle charging stations under congestion. Transportation Research Part B: Methodological 200, 103291.

MirHassani, S.A., Ebrazi, R., 2013. A Flexible Reformulation of the Refueling Station Location Problem. Transportation Science 47, 617–628.

Shu, J., Chou, M.C., Liu, Q., Teo, C.P., Wang, I.L., 2013. Models for Effective Deployment and Redistribution of Bicycles Within Public Bicycle-Sharing Systems. Operations Research 61, 1346–1359.

Wang, X., Jiang, R., Qi, M., 2022. Facility Location with Congestion and Priority in Drone-Based Emergency Delivery. arXiv:2206.03103.

Wang, Z., Lv, D., Jia, S., Wang, K., Qu, X., 2025. Urban air mobility network design and operations strategy in an urban agglomeration. Transportation Research Part E: Logistics and Transportation Review 203, 104316.

Yıldız, B., Arslan, O., Karaşan, O.E., 2016. A branch and price approach for routing and refueling station location model. European Journal of Operational Research 248, 815–826.

Appendix A. Full Model

Appendix A.1. nonlinear model

$$\max \quad Z = \sum_{(i,j)\in\mathcal{A}^{PD}} \psi_{ij} \tag{A.1}$$

$$\text{s.t.} \quad \sum_{l\in\mathcal{L}} y_{li} \le x_i, \qquad \forall i \in \mathcal{P} \tag{A.2}$$

$$y_{li} \le x_l, \qquad \forall i \in \mathcal{P}, l \in \mathcal{L} \tag{A.3}$$

$$\sum_{s\in\mathcal{S}} u_{ls} = x_l, \qquad \forall l \in \mathcal{L} \tag{A.4}$$

$$\sum_{s\in\mathcal{S}} \rho_s u_{ls} \cdot y_{li} \cdot \sum_{j:(i,j)\in\mathcal{A}^{PD}} h_{ij} x_j = \psi_{li}, \qquad \forall (l,i) \in \mathcal{A}^{LP} \tag{A.5}$$

$$\sum_{u:(u,v)\in\mathcal{A}} \psi_{uv} = \sum_{u:(v,u)\in\mathcal{A}} \psi_{vu}, \qquad \forall v \in \mathcal{N} \tag{A.6}$$

$$\psi_{ij} \le h_{ij}, \qquad \forall (i,j) \in \mathcal{A}^{PD} \tag{A.7}$$

$$\psi_{uv} = \sum_{k\in\hat{\mathcal{E}}_{uv}} \varphi_{uv}^k, \qquad \forall (u,v) \in \mathcal{A} \tag{A.8}$$

$$\sum_{v:(u,v)\in\mathcal{A}} \psi_{uv} \le \bar{\Lambda}(\mu_u) x_u, \qquad \forall u \in \mathcal{P}\cup\mathcal{D} \tag{A.9}$$

$$\sum_{(u,v)\in\mathcal{A}} \varphi_{uv}^k \le \bar{\Lambda}(\mu_k) x_k, \qquad \forall k \in \mathcal{E} \tag{A.10}$$

$$\sum_{s\in\mathcal{S}} \frac{\rho_s}{1-\rho_s} u_{ls} \le \tau \sum_{i:(l,i)\in\mathcal{A}^{L^{\mathrm{v}}P}} \psi_{li}, \qquad \forall l \in \mathcal{L}^{\mathrm{v}} \tag{A.11}$$

$$\lambda_u = \sum_{v:(u,v)\in\mathcal{A}} \psi_{uv}, \qquad \forall u \in \mathcal{P}\cup\mathcal{D} \tag{A.12}$$

$$\omega\Big(\sum_{l\in\mathcal{L}^{\mathrm{v}}}\sum_{s\in\mathcal{S}} \frac{\rho_s}{1-\rho_s} u_{ls} + \sum_{u\in\mathcal{P}\cup\mathcal{D}} \frac{\lambda_u}{\mu_u-\lambda_u} + \sum_{(u,v)\in\mathcal{A}}\sum_{k\in\hat{\mathcal{E}}_{uv}} \varphi_{uv}^k t_{uv}^k\Big)$$
$$\le (1-\varepsilon)\Omega\Big(\sum_{(u,v)\in\mathcal{A}}\sum_{k\in\mathcal{E}_{uv}} \varphi_{uv}^k + \sum_{(j,l)\in\mathcal{A}^{DL}} \varphi_{jl}^0\Big) \tag{A.13}$$

$$\sum_{l\in\mathcal{L}}\sum_{s\in\mathcal{S}} \frac{\rho_s}{1-\rho_s} u_{ls} + \sum_{u\in\mathcal{P}\cup\mathcal{D}} \frac{\lambda_u}{\mu_u-\lambda_u} + \sum_{(u,v)\in\mathcal{A}}\sum_{k\in\hat{\mathcal{E}}_{uv}} \varphi_{uv}^k t_{uv}^k \le \Gamma \tag{A.14}$$

$$\sum_{u\in\mathcal{N}} f_u x_u + c\Gamma \le C_{\text{budget}} \tag{A.15}$$

$$\Gamma \le V \sum_{l\in\tilde{\mathcal{L}}} x_l \tag{A.16}$$

$$x_u \in \{0,1\}, \quad u_{ls} \in \{0,1\}, \quad y_{li} \in [0,1]$$

$$\psi_{uv} \ge 0, \quad \varphi_{uv}^k \ge 0, \quad \lambda_u \ge 0, \quad \Gamma \ge 0$$

Appendix A.2. MISOCP

$$\max \quad Z = \sum_{(i,j)\in\mathcal{A}^{PD}} \psi_{ij} \tag{A.17}$$

$$\text{s.t.} \quad \sum_{l\in\mathcal{L}} y_{li} \le x_i, \qquad \forall i \in \mathcal{P} \tag{A.18}$$

$$y_{li} \le x_l, \qquad \forall i \in \mathcal{P}, l \in \mathcal{L} \tag{A.19}$$

$$\sum_{s\in\mathcal{S}} u_{ls} = x_l, \qquad \forall l \in \mathcal{L} \tag{A.20}$$

$$w_{lij} \le y_{li}, \quad w_{lij} \le x_j, \qquad \forall l \in \mathcal{L}, (i,j) \in \mathcal{A}^{PD} \tag{A.21}$$

$$w_{lij} \ge y_{li} - (1 - x_j), \quad w_{lij} \ge 0, \qquad \forall l \in \mathcal{L}, (i,j) \in \mathcal{A}^{PD} \tag{A.22}$$

$$D_{li} = \sum_{j:(i,j)\in\mathcal{A}^{PD}} h_{ij} w_{lij}, \qquad \forall l \in \mathcal{L}, i \in \mathcal{P} \tag{A.23}$$

$$z_{lis} \le D_{li}, \quad z_{lis} \le H_i u_{ls}, \qquad \forall l \in \mathcal{L}, i \in \mathcal{P}, s \in \mathcal{S} \tag{A.24}$$

$$z_{lis} \ge D_{li} - H_i(1 - u_{ls}), \quad z_{lis} \ge 0, \qquad \forall l \in \mathcal{L}, i \in \mathcal{P}, s \in \mathcal{S} \tag{A.25}$$

$$\psi_{li} = \sum_{s\in\mathcal{S}} \rho_s z_{lis}, \qquad \forall l \in \mathcal{L}, i \in \mathcal{P} \tag{A.26}$$

$$\sum_{u:(u,v)\in\mathcal{A}} \psi_{uv} = \sum_{u:(v,u)\in\mathcal{A}} \psi_{vu}, \qquad \forall v \in \mathcal{N} \tag{A.27}$$

$$\psi_{ij} \le h_{ij}, \qquad \forall (i,j) \in \mathcal{A}^{PD} \tag{A.28}$$

$$\psi_{uv} = \sum_{k\in\hat{\mathcal{E}}_{uv}} \varphi^k_{uv}, \qquad \forall (u,v) \in \mathcal{A} \tag{A.29}$$

$$\sum_{v:(u,v)\in\mathcal{A}} \psi_{uv} \le \bar{\Lambda}(\mu_u) x_u, \qquad \forall u \in \mathcal{P} \cup \mathcal{D} \tag{A.30}$$

$$\sum_{(u,v)\in\mathcal{A}} \varphi^k_{uv} \le \bar{\Lambda}(\mu_k) x_k, \qquad \forall k \in \mathcal{E} \tag{A.31}$$

$$\sum_{s\in\mathcal{S}} \frac{\rho_s}{1-\rho_s} u_{ls} \le \tau \sum_{i:(l,i)\in\mathcal{A}^{LvP}} \psi_{li}, \qquad \forall l \in \mathcal{L}^{\mathrm{v}} \tag{A.32}$$

$$\lambda_u = \sum_{v:(u,v)\in\mathcal{A}} \psi_{uv}, \qquad \forall u \in \mathcal{P} \cup \mathcal{D} \tag{A.33}$$

$$\left\| \begin{pmatrix} 2\sqrt{\mu_u} \\ \xi_u + \lambda_u - \mu_u + 1 \end{pmatrix} \right\|_2 \le \xi_u - \lambda_u + \mu_u + 1, \qquad \forall u \in \mathcal{P} \cup \mathcal{D} \tag{A.34}$$

$$\omega\Big(\sum_{l\in\mathcal{L}^{\mathrm{v}}} \sum_{s\in\mathcal{S}} \frac{\rho_s}{1-\rho_s} u_{ls} + \sum_{u\in\mathcal{P}\cup\mathcal{D}} \xi_u + \sum_{(u,v)\in\mathcal{A}} \sum_{k\in\hat{\mathcal{E}}_{uv}} \varphi^k_{uv} t^k_{uv} \Big)$$
$$\le (1-\varepsilon)\Omega\Big(\sum_{(u,v)\in\mathcal{A}} \sum_{k\in\hat{\mathcal{E}}_{uv}} \varphi^k_{uv} + \sum_{(j,l)\in\mathcal{A}^{DL}} \varphi^0_{jl} \Big) \tag{A.35}$$

$$\sum_{l\in\mathcal{L}} \sum_{s\in\mathcal{S}} \frac{\rho_s}{1-\rho_s} u_{ls} + \sum_{u\in\mathcal{P}\cup\mathcal{D}} \xi_u + \sum_{(u,v)\in\mathcal{A}} \sum_{k\in\hat{\mathcal{E}}_{uv}} \varphi^k_{uv} t^k_{uv} \le \Gamma \tag{A.36}$$

$$\sum_{u\in\mathcal{N}} f_u x_u + c\Gamma \le C_{\text{budget}} \tag{A.37}$$

$$\Gamma \le V \sum_{l\in\tilde{\mathcal{L}}} x_l \tag{A.38}$$

$$x_u \in \{0,1\}, \quad u_{ls} \in \{0,1\}, \quad y_{li} \in [0,1]$$
$$w_{lij} \geq 0, \quad D_{li} \geq 0, \quad z_{lis} \geq 0$$
$$\psi_{uv} \geq 0, \quad \varphi_{uv}^{k} \geq 0, \quad \lambda_u \geq 0, \quad \xi_u \geq 0, \quad \Gamma \geq 0$$

Appendix B. Parameter settings